GROUP 03: Effect of Image Scaling and Zoom Augmentation

WEEK 1

OBJECTIVE FOR WEEK 1 : Prepare the baseline dataset (YOLO format, annotated and split) and train the baseline model with no augmentation on YOLOv8n, YOLOv11n, and YOLOv26n to establish baseline performance metrics.

WORK COMPLETED:

- Downloaded and prepared the vehicle dataset (raw images + annotations).
- Consolidated dataset into a single pool and merged image - label pairs.
- Verified and defined class names against the dataset's class list.
- Verified annotations were correctly formatted in YOLO style and matched image files.
- Created a clean split Train - 70% , Val - 15%, Test - 15%.
- Generated the data.yaml configuration file for training**.**
- Configured a baseline training setup with no augmentation.
- Trained baseline model on YOLOv8n, YOLOv11n, and YOLOv26n.
- Evaluated all three models on the test split.
- Compiled baseline metrics (Precision, Recall, mAP50, mAP50-95) into a comparison table.

RESULTS:

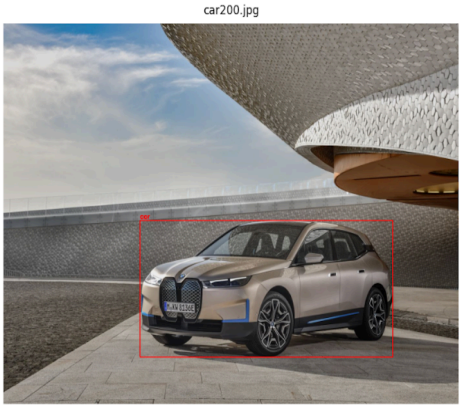
1.Dataset Details

- Sourced from Kaggle (Vehicle Dataset for YOLO) with road/traffic images annotated for 6 vehicle categories.
- Number of classes: 6.
- Class labels: 0-car, 1-threewheel, 2-bus, 3-truck, 4-motorbike, 5-van.
- Total images: 3,000.
- Train / Val / Test split: 2,100 / 450 / 450 (70% / 15% / 15%).
- Annotation format: YOLO format (normalized class : x_center, y_center, width height).
- Image sizes vary across the dataset (240–8192 px width, 151–5461 px height, avg ~740×528), most common sizes are 320×240, 500×375, and 640×480.
- All images resized to 640×640 during training (Handled by YOLO’s resizing).

- Class distribution is fairly balanced (roughly 14–18% per class), consistent across train, val, and test splits.

2.Bounding Boxes Visualisation













3.Baseline Metrics

Comparison Table for 3 Yolo versions:

YOLO Version	Precision	Recall	mAP50	mAP50-95
YOLOv8n	0.9267	0.8851	0.9329	0.8337
YOLOv11n	0.9131	0.8735	0.9189	0.8038
YOLOv26n	0.937	0.8469	0.9176	0.8283

- All three models score well above 0.91 on mAP50, so the baseline is solid across versions.
- YOLOv8n has the best recall (0.8851) and the best mAP50-95 (0.8337), it's catching the most vehicles and localizing them most accurately overall.
- YOLOv26n has the highest precision (0.937), meaning its predictions are rarely wrong, but its recall (0.8469) is the lowest, it's missing more vehicles than the others.
- YOLOv11n falls in the middle on most metrics without leading in any single one.
- Differences between the three are fairly small at this stage, so no version has a clear advantage yet.

ISSUES/CHALLENGES:

- The downloaded dataset didn't follow a single fixed folder structure , the Kaggle zip layout wasn't guaranteed to match our assumptions, so we had to manually inspect the folder tree before trusting any file paths, rather than hardcoding them upfront.
- Some images didn't sit in a straightforward images/labels sibling folder pairing, so a fallback search was added to locate matching label files anywhere in the raw dataset if the default path guess failed.
- The source dataset only came with train and validation folders, no test split. We pooled all the data ourselves and created a fresh 70/15/15 train/val/test split so we'd have a proper held-out test set for evaluation.
- We weren't initially sure how many epochs to train for, so we ran YOLOv8n first as a test to gauge training time and performance before settling on 30 epochs for all three models. This caused the YOLOv8n run to be executed more than once,

and Ultralytics auto-renamed the output folder (baseline_yolov8n-2) as a result, we had to be careful to reference the correct run folder when pulling final results.

PLAN FOR NEXT WEEK:

- Define Low/Medium/High intensity levels for the Scaling/Zoom augmentation with justification for chosen values.
- Implement the augmentation in code on top of the existing baseline pipeline.
- Generate and visually inspect before/after sample grids for each intensity level.
- Document everything in the Week 2 report.

HOURS SPENT (PER MEMBER):

Members	Hours Spent	Contribution
Simran Koshta (B230707CS)	3	Dataset download, unzip, and folder structure inspection,Report.
Gunna Mohana Prathyusha (B230324CS)	3.5	Label verification, class name mapping, and image-label pairing into a single pool,Report.
Maithili Kolasani (B230385CS)	3.5	Bounding box visualisation and sanity checks,Report.
Ardhra Alaina Winston (B230198CS)	3	Train/val/test split creation and data.yaml setup,Report.
Sreeneha Bhavanasi (B230092CS)	4.5	Baseline training for all three YOLO versions (YOLOv8n, YOLOv11n, YOLOv26n) and metrics evaluation,Report.